

# Multi-Trait Models

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Multi-trait Bayesian linear regression (MT-BLR) models several traits jointly so that evidence can be shared through marker-effect patterns and covariance structures. This note introduces the main model ideas and explains how they relate to the current `sblr` implementation. The deeper requirements for cross-trait summary statistics and sample overlap are covered in [Technical Appendix: Multi-Trait Overlap](#).

# 1 Why Model Traits Jointly

Biologically related traits often share part of their genetic architecture. A marker with a weak effect on one trait may be supported by stronger evidence from another trait, and a joint model can estimate genetic and residual covariances rather than treating every trait as independent.

The main reasons to use an MT-BLR model are:

- to borrow information across correlated traits;
- to represent markers that affect one, several, or no traits;
- to estimate cross-trait marker-effect, genetic, and residual covariance;
- to improve inference for weak signals when the assumed sharing structure is appropriate.

This is different from fitting a single-trait model independently to every column of a phenotype or summary-statistics matrix. Parallel single-trait fits can be useful and computationally convenient, but they do not jointly update marker effects or estimate cross-trait covariance during sampling.

# 2 Multi-Trait Regression Model

For  $t$  traits measured on a common set of  $n$  individuals, the individual-level model is

$$\mathbf{Y} = \mathbf{XB} + \mathbf{E},$$

where:

- $\mathbf{Y}$  is the  $n \times t$  phenotype matrix;
- $\mathbf{X}$  is the  $n \times m$  genotype matrix;
- $\mathbf{B}$  is the  $m \times t$  marker-effect matrix;
- $\mathbf{E}$  is the  $n \times t$  residual matrix.

The  $i$ th row of  $\mathbf{B}$ ,

$$\mathbf{b}_i = [b_{i1} \quad \cdots \quad b_{it}]^\top,$$

is the effect vector for marker  $i$ . A marker may have no effect, affect one trait, or affect several traits. Joint MT-BLR models use prior probabilities and covariance matrices to describe these possibilities.

### 3 Marker Effects, Sharing, and Pleiotropy

A multi-trait marker model needs to distinguish between two related ideas:

1. **Trait-sharing pattern:** which elements of  $\mathbf{b}_i$  are active.
2. **Effect covariance:** how the non-zero effects vary together across traits.

For three traits, example sharing patterns are:

| Pattern | Interpretation                        |
|---------|---------------------------------------|
| 000     | Marker is inactive for all traits     |
| 100     | Marker is active only for trait 1     |
| 110     | Marker is active for traits 1 and 2   |
| 111     | Marker is active for all three traits |

The model can assign prior probabilities to these patterns and update their posterior support. A shared active pattern is compatible with pleiotropy, but it is not proof that one marker is the biological cause of multiple traits. LD, prior assumptions, trait scaling, and the available summary statistics also influence the inferred sharing pattern.

### 4 Multi-Trait Sparse Priors

The single-trait sparse prior combines an exact null component with an active normal component. MT-BLR generalizes this idea by assigning each marker a trait-sharing model  $s_i$ . Conditional on  $s_i$ , inactive effects are fixed at zero and active effects are drawn from a multivariate distribution.

A schematic prior is

$$s_i \sim \text{Categorical}(\pi),$$

$$\mathbf{b}_{i,s_i} \mid s_i \sim \mathcal{N}(\mathbf{0}, \Sigma_{b,s_i}),$$

where  $\pi$  contains prior probabilities for the allowed sharing patterns and  $\Sigma_{b,s_i}$  is the active-effect covariance submatrix for pattern  $s_i$ .

The allowed model space is an important modeling decision:

- A full model space permits all  $2^t$  inactive/active patterns.
- A restrictive model space may permit only all-null and all-active patterns.
- Custom prior probabilities can favor trait-specific or broadly shared effects.

As the number of traits increases, the full  $2^t$  model space grows quickly. Restricting or regularizing the sharing patterns can therefore be both a scientific choice and a computational necessity.

The documented `sblr()` interface exposes sharing-model inputs such as `models` and `pimodels`. These describe the permitted patterns and their prior probabilities. Exact argument support and behavior should be checked for the selected experimental algorithm before a production analysis.

## 5 Genetic, Effect, and Residual Covariance

Covariance matrices are central to multi-trait inference.

### 5.1 Active-effect covariance

The active-effect covariance,  $\Sigma_b$ , describes how non-zero marker effects vary together across traits. A positive off-diagonal entry favors effects with the same direction; a negative entry favors opposite directions. This covariance is conditional on the marker being active for the corresponding traits.

### 5.2 Genetic covariance

The aggregate genetic value is

$$\mathbf{G}_{\text{value}} = \mathbf{X}\mathbf{B}.$$

Its covariance across traits depends on both the fitted marker effects and LD. Genetic covariance therefore summarizes the combined architecture rather than only the conditional covariance of active effects.

### 5.3 Residual covariance

Residual vectors are commonly modeled as

$$\mathbf{e}_r \sim \mathcal{N}(\mathbf{0}, \Sigma_e),$$

where  $\Sigma_e$  captures cross-trait association not explained by the fitted marker effects. Environmental effects, unmodeled genetic effects, measurement processes, and shared individuals can all contribute.

Effect, genetic, and residual covariance answer different questions and should not be interpreted interchangeably.

## 6 Individual-Level and Summary-Statistics MTBLR

At the individual level, joint MT-BLR uses  $\mathbf{X}$  and  $\mathbf{Y}$  directly. With complete data, it can calculate all required within-trait and cross-trait products from the observed individuals.

At the summary-statistics level, the same calculations must be represented by compatible cross-products:

$$\mathbf{X}^\top \mathbf{Y}, \quad \mathbf{X}^\top \mathbf{X}, \quad \mathbf{Y}^\top \mathbf{Y}.$$

These correspond to:

- marker-by-trait cross-products, often denoted  $\mathbf{Xy}$  or  $\mathbf{wy}$ ;
- marker cross-products or LD;
- the trait cross-product matrix  $\mathbf{SSY}$ ;
- trait-specific sample sizes and, where needed, pairwise sample-overlap information.

For one trait,  $y^\top y$  is a scalar. For several traits,  $\mathbf{Y}^\top \mathbf{Y}$  contains both trait-specific sums of squares and cross-trait products. The off-diagonal terms are necessary for estimating and updating residual covariance.

The summary statistics, LD, sample sizes, trait order, marker order, and scaling assumptions must all be compatible. See [Technical Appendix: Summary Statistics](#) for the cross-product representation and [Technical Appendix: Multi-Trait Overlap](#) for the role of  $\mathbf{SSY}$  and sample overlap.

## 7 Sample Overlap in Practice

Full sample overlap is the simplest setting: all traits are measured on the same individuals, and  $\mathbf{SSY}$  can be computed directly from the phenotype matrix. Cross-trait residual covariance then has a direct interpretation on that shared sample.

With partial overlap, each trait pair may share a different number of individuals. A pairwise overlap matrix, often denoted  $\mathbf{N}_{\text{mat}}$ , records those counts. Cross-trait phenotype products or covariances then require explicit assumptions about the shared samples and trait scaling.

The current CSR multi-trait implementation is closest to a shared-LD, equal-sample-size setting. It consumes cross-trait phenotype information but does not expose a general  $\mathbf{N}_{\text{mat}}$ -aware workflow for arbitrary partial overlap. Analyses with partial or uncertain overlap should therefore be treated as a technical extension, not as a routine current workflow.

## 8 Simulation and Model Development

Simulation is useful for checking whether a proposed MT-BLR model can recover known sharing and covariance structures.

`mtsim()` simulates multiple traits with shared and trait-specific causal markers, target heritabilities, and target genetic and residual correlations. A short schematic call is:

```
sim <- mtsim(
  W = W,
  nt = 3,
  n_shared = 30,
  n_specific = 10,
  h2 = c(0.4, 0.5, 0.3),
  rg = target_rg,
  re = target_re,
  seed = 1
)
```

Simulation output can be used to compare target and realized covariance, inspect true marker-effect patterns, and assess recovery under different priors or LD approximations.

`mtsim_annotation()` extends this development setting with overlapping annotations and annotation-informed marker architectures. `summarize_annotation_signal()` can summarize the simulated signal. The current polished annotation workflows primarily fit single-trait or parallel single-trait CSR models; annotation-aware joint MT-BLR remains a research direction. See [Annotation-Informed Models](#).

## 9 Outputs and Interpretation

Multi-trait fits may return the following objects, depending on the selected sampler:

- **bm**: posterior mean marker effects, with markers in rows and traits in columns;
- **dm**: posterior inclusion summaries or PIPs by marker and trait;
- **d** or **o**: current inclusion or sharing-model states, when returned;
- **vbs**, **vgs**, and **ves**: sampled effect, genetic, and residual variance summaries;
- **covb**, **covg**, and **cove**: posterior effect, genetic, and residual covariance summaries;
- **rb**, **rg**, and **re**: corresponding correlation summaries, when returned;
- **pi** or **pim**: posterior inclusion or sharing-model probability summaries.

Useful reporting separates trait-specific and cross-trait evidence:

- summarize PIPs and posterior mean effects by trait;
- list top markers separately for each trait;
- identify markers with support across several traits;
- report covariance and correlation estimates with uncertainty;
- compare posterior sharing probabilities with their prior values.

Large marker-by-trait matrices should usually be summarized rather than printed. Posterior support is conditional on the model, priors, LD, cross-trait information, and MCMC settings. A high multi-trait PIP or shared pattern identifies a candidate under those assumptions; it does not establish causal direction.

## 10 Annotation-Informed Multi-Trait Directions

Annotations can, in principle, modify the prior probability of each trait-sharing pattern, the scale of active effects, or the covariance among traits. This creates several possible MT-BLR extensions:

- marker-specific probabilities for trait-specific and shared patterns;
- annotation-dependent active-effect variance or covariance;
- group-specific sharing priors;
- annotation-dependent mixture components.

These extensions require careful regularization because overlapping annotations, many sharing patterns, and covariance estimation can create a large parameter space. In the current package, annotation simulations and annotation-informed single-trait CSR wrappers provide the main implemented foundation. They should not be described as a polished annotation-aware joint MTBLR workflow.

## 11 When to Use Multi-Trait Models

Consider a multi-trait model when:

- traits are biologically or statistically related;
- shared genetic architecture is scientifically meaningful;
- trait scaling and phenotype definitions are compatible;
- sample overlap is known or defensibly modeled;
- the LD reference and marker ordering are compatible across traits;
- there is enough information to estimate cross-trait covariance.

Prefer separate single-trait analyses, or use them as a diagnostic comparison, when trait definitions are incompatible, sample overlap is unknown, LD references differ materially, or cross-trait borrowing is not scientifically justified.

## 12 Practical Cautions and Diagnostics

Multi-trait models introduce more assumptions than single-trait models. Important checks include:

- verify trait order across  $\mathbf{X}_y$ ,  $\mathbf{SSY}$ , sample sizes, covariance matrices, and output columns;
- verify marker order and scaling across all trait summary statistics and LD;
- check that  $\mathbf{SSY}$  and covariance matrices are symmetric and numerically plausible;
- record sample-overlap assumptions;
- compare joint results with separate single-trait fits;
- inspect covariance traces and sharing-model probabilities;
- assess sensitivity to sharing priors and covariance priors;
- use longer chains, repeated chains, and convergence diagnostics for production inference.

Weak or unstable covariance estimates can lead to misleading borrowing across traits. Short demonstration chains are useful for integration testing, not for final scientific conclusions.

## 13 Current Status in `sblr`

The package contains a conceptual and implementation foundation for MT-BLR:

- `sblr()` is the documented high-level interface for several experimental summary-statistics algorithms and exposes sharing-model and covariance inputs;

- native multi-trait implementations explore dense, Armadillo-based, OpenMP, and CSR computations;
- the CSR multi-trait kernel supports cross-trait phenotype information and shared sparse LD under relatively restrictive sample-size and scaling assumptions;
- `mtsim()` and `mtsim_annotation()` support development and model checking.

The most complete current workflow examples focus on BED and CSR single-trait models, including annotation-informed extensions. There is not yet an equally polished, general, end-to-end public workflow for joint MTBLR with arbitrary sample overlap. Algorithm-specific argument contracts and returned objects should be validated before relying on an experimental MTBLR path.

This status distinction matters: the package contains real multi-trait model and sampler components, but some interfaces and broader overlap or annotation extensions remain research-stage.

## 14 Related Notes and Sources

- [Single-Trait Models](#) explains the sparse single-trait model that MT-BLR generalizes.
- [Technical Appendix: Multi-Trait Overlap](#) gives the detailed treatment of `SSY`, `Nmat`, and overlap assumptions.
- [Technical Appendix: Summary Statistics](#) derives the summary-statistics representation.
- [Data Representations](#) compares individual-level and summary-statistics inputs.
- [Annotation-Informed Models](#) describes the currently exposed annotation-aware CSR models.
- [Workflows and Outputs](#) explains practical fit summaries and reporting.

This note draws on the archived `sblr_theory.qmd` draft, the completed technical appendices, the documented `sblr()` and simulation interfaces, and the current experimental multi-trait sampler implementations.